

Flood Risk Mapping of the Nag River Basin, Nagpur

A DEM-Based Hydrological Terrain Analysis Using QGIS and Python

An Independent Study

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2026

Abstract

This study develops a terrain-based flood risk map for a corridor of the Nag River basin in Nagpur, covering Ambazari Lake, Sitabuldi, and the Mor Bhawan/Variety Chowk area. Using a Digital Elevation Model (DEM) processed in QGIS, a drainage network was derived through standard hydrological terrain analysis (sink filling, flow direction, and flow accumulation), and a composite flood risk score was generated from elevation, slope, and proximity to derived drainage channels. The resulting risk map was validated against documented flooding locations from the severe Nagpur flood event of 23 September 2023. A supplementary Python script was used to quantify the spatial extent of each risk category and to statistically summarise risk composition around validation points. Results show strong model agreement at Ambazari Lake, a naturally terrain-driven flood zone, but under-prediction of risk in the densely urbanised Sitabuldi corridor — a discrepancy attributable to anthropogenic factors (encroachment, drainage infrastructure limitations) that are not captured by DEM-based terrain analysis alone. The study demonstrates both the value and the limitations of purely topographic flood-risk modelling in a rapidly urbanising Indian city.

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1. Introduction

Urban flooding has become an increasingly frequent hazard in Indian cities, driven by a combination of intense monsoon rainfall, rapid unplanned urbanisation, and the loss of natural drainage capacity. Nagpur, a rapidly growing city in Maharashtra, experienced a severe flood event on 23 September 2023, when intense rainfall overwhelmed the Nag River and its surrounding drainage network, affecting several central localities including Ambazari, Sitabuldi, Mor Bhawan, and Variety Chowk.

This study aims to develop a flood risk map for this corridor using freely available Digital Elevation Model (DEM) data and standard GIS-based hydrological terrain analysis. The objective is to identify terrain characteristics — low elevation, flat slope, and proximity to natural drainage convergence — that are associated with flood susceptibility, and to test how well such a purely terrain-driven model predicts the locations that were actually affected during the 2023 event. The project also incorporates a light Python-based analysis component to quantify the spatial results numerically.

2. Study Area

The study focuses on a corridor of the Nag River basin spanning Ambazari Lake in the west, through Sitabuldi in the city centre, to the Mor Bhawan / Variety Chowk area. This corridor was deliberately selected because it corresponds to the area most severely affected during the September 2023 Nagpur flood event, allowing the model's predictions to be validated against real, documented flood locations rather than an arbitrary or unvalidated extent.

Ambazari Lake functions as a natural regulator at the upstream end of this stretch of the Nag River, while Sitabuldi and the surrounding Mor Bhawan / Variety Chowk area represent a densely built-up, highly urbanised mid-city zone. This mix of a natural waterbody and a heavily urbanised corridor makes the study area well suited to examining both the strengths and the limitations of a purely terrain-based flood risk model.

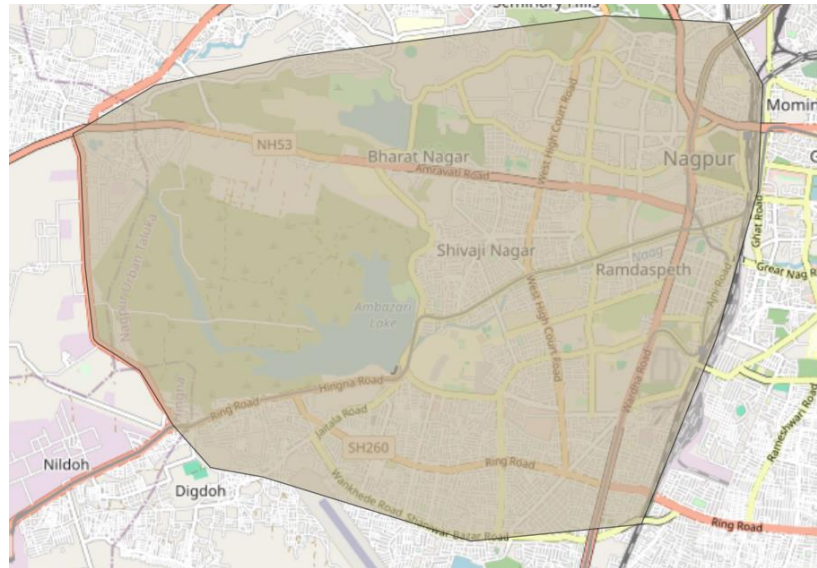


Figure 1: Study area corridor covering Ambazari Lake, Sitabuldi, and the Mor Bhawan / Variety Chowk area, overlaid on OpenStreetMap base data.

3. Data and Methodology

3.1 Data Sources

- Digital Elevation Model (DEM): CartoDEM Version 3 (Cartosat-1 derived DEM), obtained via the Bhuvan Open Data Archive
- River and drainage network reference: OpenStreetMap (via the QuickOSM plugin in QGIS)
- Flood event reference data: documented locations and rainfall data associated with the 23 September 2023 Nagpur flood event, drawn from published news reports and academic literature

CartoDEM Version 3, produced from Cartosat-1 stereo imagery, was used as the primary elevation data source for this study, offering spatial resolution and vertical accuracy well suited to city-corridor-scale flood risk analysis.

3.2 Software and Tools

- QGIS 4.2.0 — primary GIS platform for all raster and vector processing
- GRASS GIS tools (via QGIS Processing Toolbox) — `r.watershed`, for hydrologically-correct flow accumulation
- Native QGIS raster tools — Fill Sinks (Wang & Liu), Slope, Proximity (Raster Distance), Reclassify by Table, Raster Calculator
- Python (QGIS Python Console) — for quantitative post-processing and statistical summary of results

3.3 Methodology Overview

The methodology followed six stages:

1. DEM acquisition and clipping to the study area boundary, with correction of NoData values along the boundary edge
2. Reprojection of the DEM to a projected coordinate system (WGS 84 / UTM Zone 44N) to ensure hydrological tools operate in consistent real-world distance units (metres)
3. Depression (sink) filling to remove artificial pits from the DEM prior to hydrological analysis
4. Derivation of flow direction and flow accumulation using GRASS `r.watershed`, producing a drainage network based on cumulative upstream contributing area per cell
5. Generation of three risk-contributing layers — elevation, slope, and proximity to derived high-accumulation (stream) cells — each reclassified to a common 1 (low) to 3 (high) risk score, then summed into a composite flood risk score and smoothed using a sieve filter to remove isolated single-pixel noise
6. Validation of the resulting risk classification against documented flood locations from the September 2023 event, both by direct pixel sampling and by statistical summary of risk composition within a 100 m buffer around each location

This workflow follows standard DEM-based hydrological terrain analysis principles, in which surface water flow is modelled as moving from each cell toward its lowest neighbouring cell, allowing flow direction and cumulative contributing area (flow accumulation) to be derived computationally from elevation data alone.

4. Results — Drainage Network and Flow Accumulation

Flow accumulation analysis using GRASS `r.watershed` produced a clear, hydrologically coherent drainage network across the study area, with a maximum accumulation value of 4,725 cells — corresponding to points where the largest upstream catchment area converges. The resulting branching (dendritic) pattern of high-accumulation cells corresponds visually to the expected course of the Nag River and its minor tributaries through the study corridor, providing a basic visual check on the validity of the terrain-derived drainage network before it was used in risk classification.

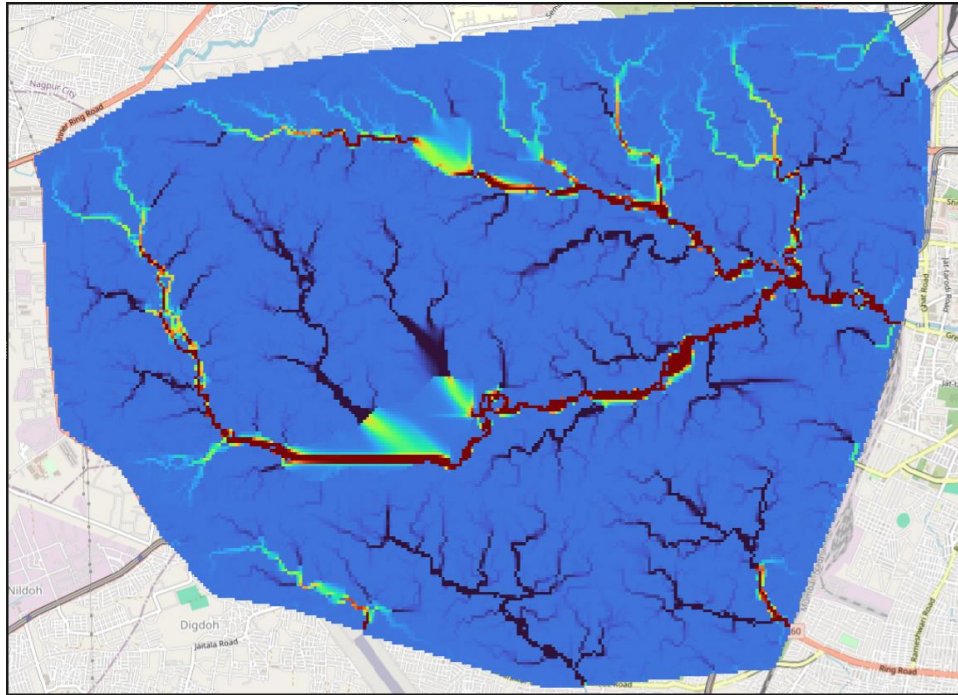


Figure 2: Derived flow accumulation raster (spectral colour ramp; red/orange = highest accumulation, forming the derived drainage network), overlaid on the study area.

Note: cells shown in the darkest tone near the western boundary (close to Ambazari Lake) carry negative accumulation values. This is a known edge effect of GRASS `r.watershed`, arising because the clipped study area boundary does not represent a complete natural watershed — some of the true upstream catchment for this area lies outside the clipped extent, so the tool cannot compute a fully confident accumulation value there. This is discussed further as a limitation in Section 8.

An initial attempt to compute flow accumulation using GRASS `r.flow` produced implausibly low accumulation values (maximum of 124, versus an expected range in the thousands for a raster of this extent). This was traced to a methodological difference: `r.flow` computes accumulation via a sampling-based approach that traces a limited number of flow lines across the terrain, rather than true per-cell cumulative contributing area. `r.watershed`, which computes genuine cumulative upstream contributing area for every cell, was used instead and produced results consistent with expected drainage behaviour.

5. Results — Flood Risk Classification

Elevation, slope, and proximity-to-drainage layers were each reclassified into a 1–3 risk score and summed to produce a composite flood risk score ranging from 3 (lowest risk) to 9 (highest risk), which was further grouped into three final categories: Low, Medium, and High risk.

5.1 The Three Risk Factor Layers

Each of the three contributing layers captures a different physical driver of flood susceptibility:

- Elevation risk: low-lying ground sits closer to the water table and to the river's natural water level and is more likely to be inundated first. Lower elevation was scored as a higher risk.

- Slope risk: flat terrain cannot drain surface water away quickly, so water pools and accumulates. Steep terrain drains rapidly and rarely floods. Flatter slope (lower degree value) was scored as a higher risk.
- Proximity risk: cells located close to the derived drainage network (the high flow-accumulation "stream" cells identified in Section 4) are the first to be affected when a channel exceeds capacity, and the risk decreases with distance from drainage. Shorter distance to the nearest derived stream cell was scored as higher risk.

Each layer was independently reclassified into three discrete classes (1 = low risk, 2 = medium risk, 3 = high risk) before being summed into the composite score described above. The exact thresholds used for each layer, calibrated to the actual data range observed within the study area, are given below.

Elevation Range	Risk Score	Risk Level
222 m – 250 m	3	High
250 m – 280 m	2	Medium
280 m – 306 m	1	Low

Table: Elevation reclassification thresholds (study area elevation range: 222–306 m).

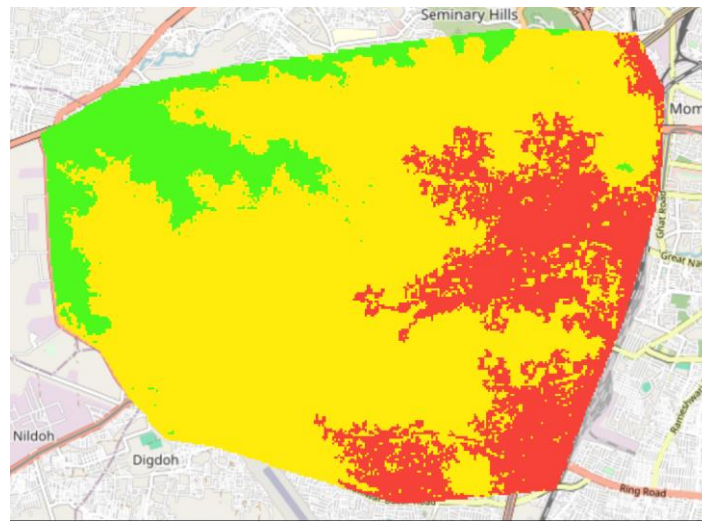


Figure 3: Elevation risk classification (green = low risk / higher ground, yellow = medium risk, red = high risk / low-lying ground). High-risk zones are concentrated in the eastern part of the corridor toward Sitabuldi and Momin Pura, and along the southern edge of the study area.

Slope Range	Risk Score	Risk Level
0° – 2°	3	High (flat)
2° – 5°	2	Medium
5° – 21.4°	1	Low (steep)

Table: Slope reclassification thresholds (study area slope range: 0°–21.4°, mean 2.67°).

Distance to Drainage	Risk Score	Risk Level
0 m – 5 m	3	High (close)
5 m – 12 m	2	Medium
12 m – 40 m	1	Low (far)

Table: Proximity-to-drainage reclassification thresholds (study area distance range: 0–39.8 m, mean 11.1 m).

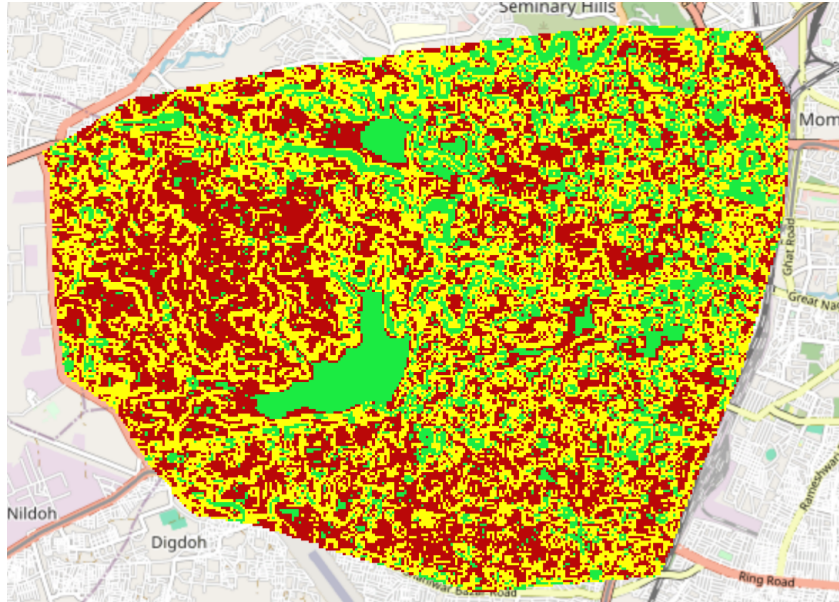


Figure 4: Slope risk classification (dark red = low risk / steep terrain, yellow = medium risk, green = high risk / flat terrain). Ambazari Lake is clearly visible as a solid green (high-risk) patch, consistent with its flat surface being correctly scored as high risk under this criterion.

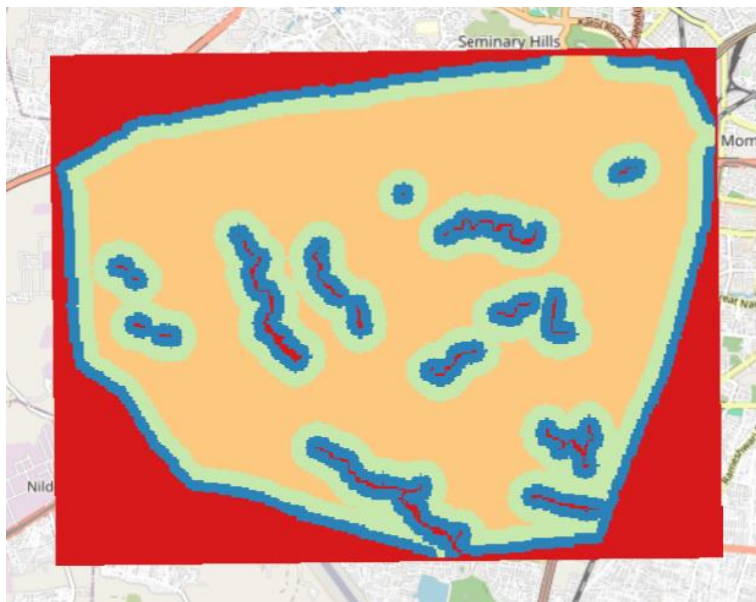


Figure 5: Proximity-to-drainage risk classification (thin red lines = high risk, directly along derived stream channels; blue = medium risk buffer; orange = low risk, far from drainage). The solid red border is a display artefact from NoData cells outside the study area boundary and does not represent real risk classification.

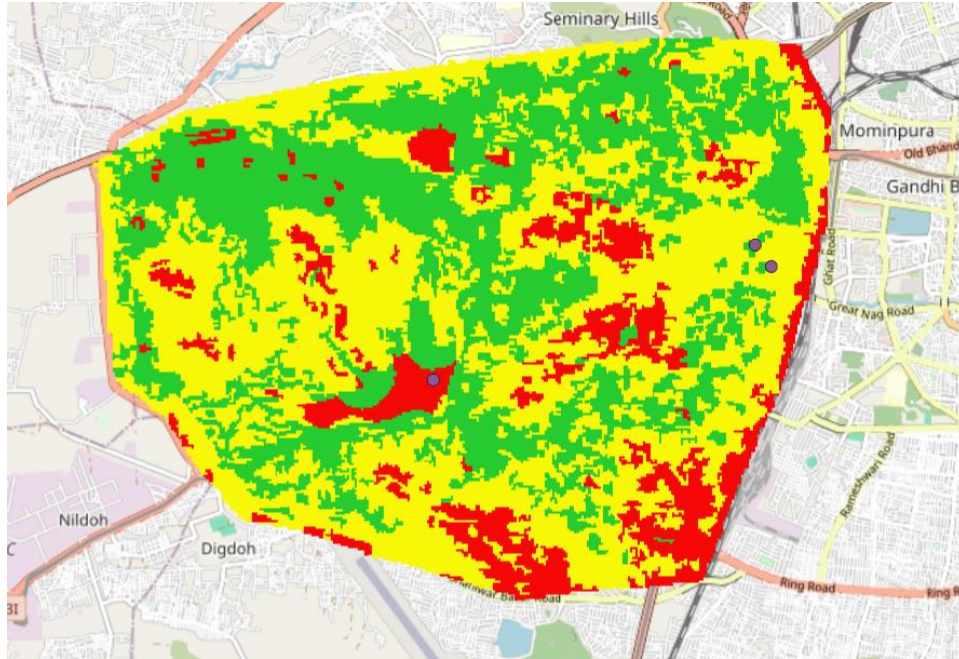


Figure 6: Final classified flood risk map (green = low risk, yellow = medium risk, red = high risk), with validation points overlaid.

A quantitative breakdown of the study area by risk category, generated via the Python script described in Section 7, is as follows:

Risk Category	Area (km ²)	Percentage of Study Area
Low Risk	17.70	36.6%
Medium Risk	24.33	50.3%
High Risk	6.37	13.2%

High-risk zones (13.2% of the study area) were concentrated in specific, spatially coherent pockets rather than distributed evenly — notably around Ambazari Lake, and along the southern and eastern peripheries of the study corridor, rather than forming a single contiguous flood-risk band.

6. Validation Against the September 2023 Flood Event

The classified risk map was validated against three documented locations affected during the 23 September 2023 Nagpur flood event: Ambazari, and two locations in the Sitabuldi / Mor Bhawan corridor. Validation was performed two ways: (a) direct sampling of the risk classification at each point's exact coordinates, and (b) statistical summary of risk composition within a 100 m buffer around each point, generated via Python (Section 7), to account for sensitivity to exact point placement at the DEM's 30 m pixel resolution.

Location	Exact Pixel Risk	100 m Buffer Composition	Assessment
Ambazari	High	100% High	Strong match
Sitabuldi	Low	64% Low, 36% Medium, 0% High	Mismatch — model under-predicts

Mor Bhawan//Variety Chowk	Low	44% Low, 56% Medium, 0% High	Mismatch, closer to boundary
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The Ambazari location, where flooding is driven primarily by natural terrain convergence around the lake, was correctly and unambiguously classified as high risk — 100% of the surrounding 100 m buffer fell within the high-risk category. This supports the validity of the terrain-based approach for naturally terrain-driven flood mechanisms.

In contrast, both Sitabuldi-area locations were classified as low risk at the exact point, with no high-risk pixels present anywhere within their 100 m buffers. This represents a genuine under-prediction by the model, not a marginal or boundary-sensitive artefact — the buffer statistics confirm the surrounding neighbourhood is genuinely dominated by low-to-medium risk classification, not merely a single mis-classified pixel.

This pattern is consistent with the documented drivers of the 2023 Nagpur flood. Published analyses of the event report that encroachment reduced the extent of natural water retention areas by approximately 81%, while built-up area within the city nearly tripled between 2000 and 2023, substantially reducing natural drainage capacity in exactly the corridor where the model's predictions fell short. These are anthropogenic, infrastructure-driven factors — not captured by a DEM, which records only bare-earth terrain elevation. The contrast between strong model performance at Ambazari (a natural water body) and under-prediction at Sitabuldi (a dense urban corridor) is itself a meaningful and expected finding: it demonstrates that DEM-derived terrain models are most reliable for naturally terrain-driven flood mechanisms, and require supplementary land-use, impervious-surface, and drainage-infrastructure data to adequately represent flood risk in dense urban settings.

7. Python-Based Quantitative Analysis

A supplementary Python script was developed and executed within the QGIS Python Console to quantify results numerically rather than relying solely on visual map interpretation. Two analyses were performed:

- Area/percentage summary: reading the classified risk raster as a numeric array (via NumPy) and calculating the real-world area (km²) and percentage share of each risk category across the study area (results presented in Section 5).
- Buffer-zone validation statistics: using QGIS's Processing framework (accessed via Python) to generate 100 m buffer zones around each flood validation point, then computing a zonal histogram of risk-category pixel counts within each buffer, alongside direct point-sampling of the risk raster at each location's exact coordinates (results presented in Section 6).

This component demonstrates the ability to combine GIS processing algorithms with custom Python scripting to move from a purely visual/qualitative GIS output toward a quantified, statistically defensible result — a workflow directly applicable to further geoinformatics and data-driven civil engineering work.

8. Discussion and Limitations

This study demonstrates that a DEM-based terrain analysis, using only elevation, slope, and derived drainage proximity, can meaningfully identify flood-prone terrain in areas where flooding is primarily a function of natural topography — as shown by the strong validation result at Ambazari Lake. However, the validation results also reveal a clear and important limitation: in the densely urbanised Sitabuldi corridor, the model under-predicted risk at both validated locations, with no high-risk classification present even within a 100 m neighbourhood of either point.

This limitation is not a flaw in the implementation, but an inherent characteristic of purely terrain-based flood modelling. A DEM captures only bare-earth surface elevation; it cannot represent impervious surface cover, stormwater drainage capacity, encroachment on natural water channels, or other anthropogenic modifications to how water moves through a city — all of which were documented as significant contributors to the September 2023 Nagpur flood, particularly in its urbanised core. This underscores a well-recognised principle in urban flood risk assessment: terrain analysis is a necessary but not sufficient basis for flood risk mapping in built-up areas, and should ideally be

supplemented with land-use/land-cover classification, impervious surface mapping, and drainage-network infrastructure data for more complete urban flood risk assessment.

Other limitations of this study include: a relatively small number of validation points (three, limited to locations with clearly documented 2023 flood impacts), and reliance on elevation-derived drainage rather than the true, engineered stormwater drainage network of the city, which does not necessarily follow natural terrain flow paths in a heavily modified urban area.

9. Conclusion

This study successfully developed and validated a DEM-based flood risk map for a corridor of the Nag River basin in Nagpur, using standard GIS hydrological terrain analysis combined with a light Python-based quantitative validation component. The model correctly identified high flood risk at Ambazari Lake, a naturally terrain-driven flood zone, achieving 100% high-risk classification within its validation buffer. However, it under-predicted risk in the densely urbanised Sitabuldi corridor, reflecting the well-documented role of encroachment and inadequate drainage infrastructure — rather than terrain alone — in driving the September 2023 Nagpur flood in that area.

These findings collectively demonstrate both the genuine value and the clear limitations of terrain-based flood risk modelling, and point toward a practical direction for future work: integrating land-use, impervious-surface, and drainage-infrastructure data with DEM-based terrain analysis to produce a more complete urban flood risk assessment for rapidly growing Indian cities such as Nagpur.

10. References

Nagpur flood event data and contributing factors (September 2023) — drawn from published news coverage and academic analysis of the event, including a Scientific Reports study evaluating flood dynamics and land-use change in Nagpur city using remote sensing and GIS.

CartoDEM Version 3 (Cartosat-1 DEM) — National Remote Sensing Centre (NRSC), Indian Space Research Organisation (ISRO), Bhuvan Open Data Archive.

QGIS Development Team — QGIS Geographic Information System, and the GRASS GIS watershed hydrological modelling tool.